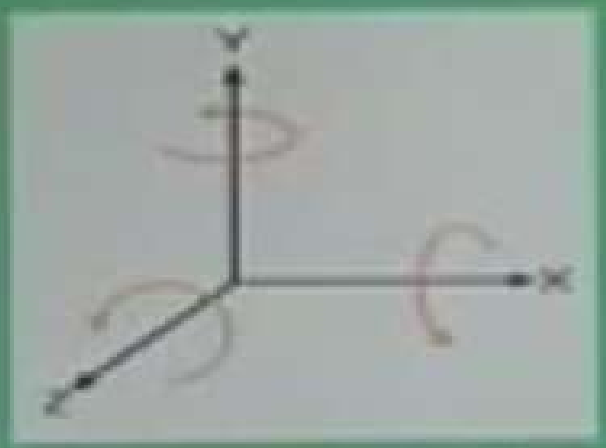
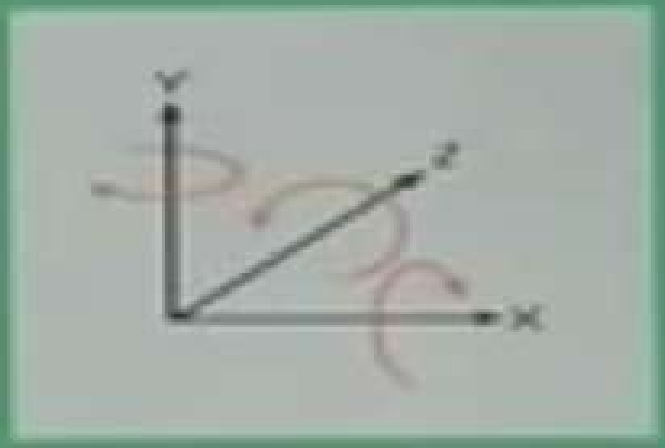
Three-Dimensional Graphics

- Use of a right-handed coordinate syste
(consistent with math)
- Left-handed suitable to screens.
- To transform from right to left, negate th
z values.



Right Handed Space



Left Handed Spac

Homogeneous representation of point in 3D space:

$$P = | x\ y\ z\ w |^T$$

$$(w = 1, \text{for a 3D point})$$

Transformations will thus be represented by 4x4 matrices:

$$P' = A.P$$

Transformation Matrix in 3D:

$$A = \begin{bmatrix} a & b & c & p \\ d & e & f & q \\ g & i & j & r \\ l & m & n & s \end{bmatrix} = \begin{bmatrix} \mathrm{T} & \mathrm{K} \\ \Gamma & \Theta \end{bmatrix}$$

where,

$$\mathrm{T} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & i & j \end{bmatrix}$$

produces linear transformatio
scaling, shearing, reflection
and rotation.

$K = [p\ q\ r]^T$, produces transla

$\Gamma = [l\ m\ n]^T$, yields perspective transformati

while, $\Theta = s$, is responsible for uniform sca

$$\begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Translation

$$\begin{bmatrix} Sx & 0 & 0 & 0 \\ 0 & Sy & 0 & 0 \\ 0 & 0 & Sz & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Scale

$$\begin{bmatrix} 1 & 0 & Sh_x & 0 \\ 0 & 1 & Sh_y & 0 \\ 0 & Sh_z & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Shear

Origin is unaffected by scale and shear

3D Reflection:

The following matrices:

$$T_{XY} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T_{YZ} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T_{ZX} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

produce reflection about:

XY plane

YZ plane

ZX plane

respectively.

Rotation Matrices along an axis:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) & 0 \\ 0 & \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

X-axis

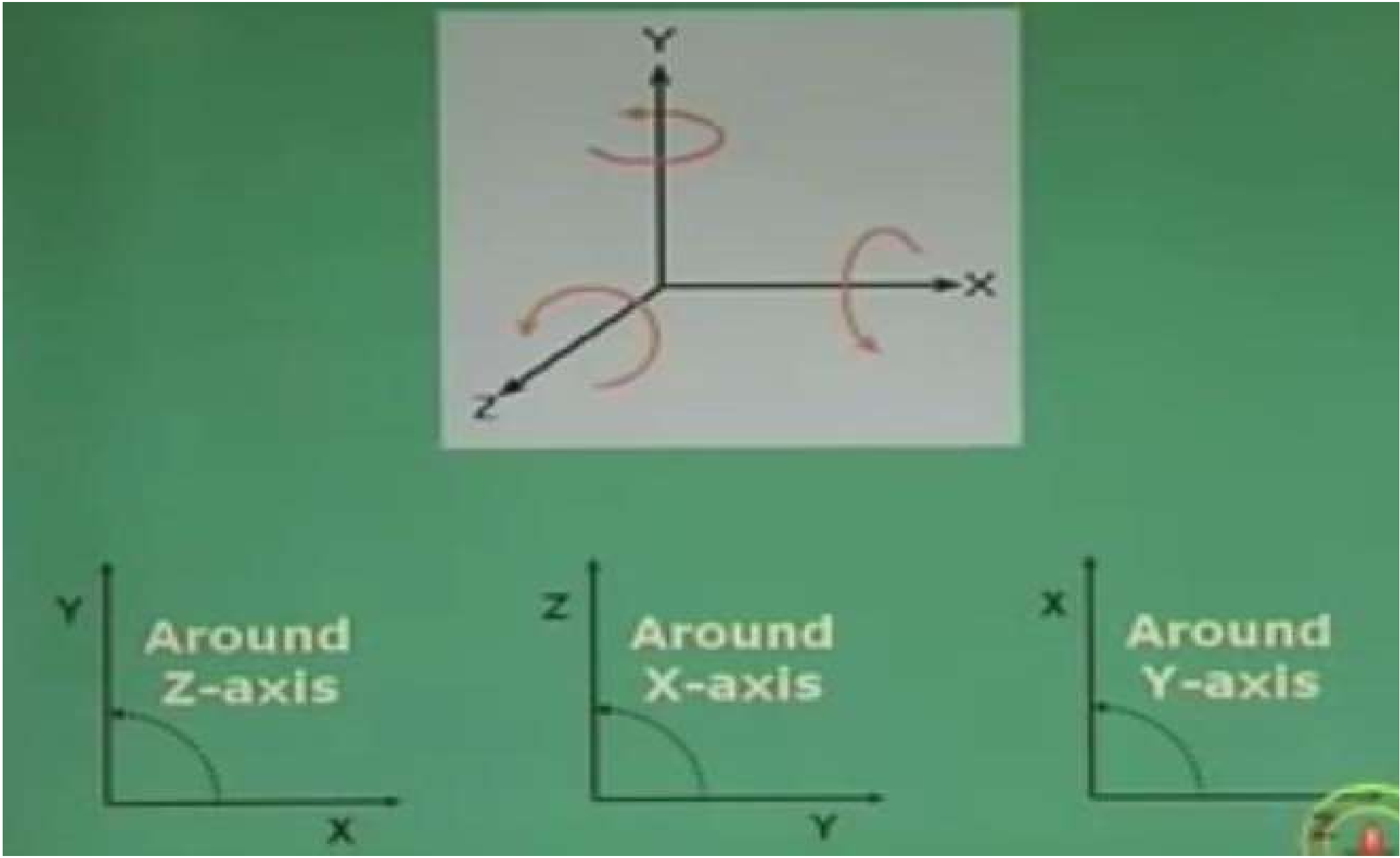
$$\begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

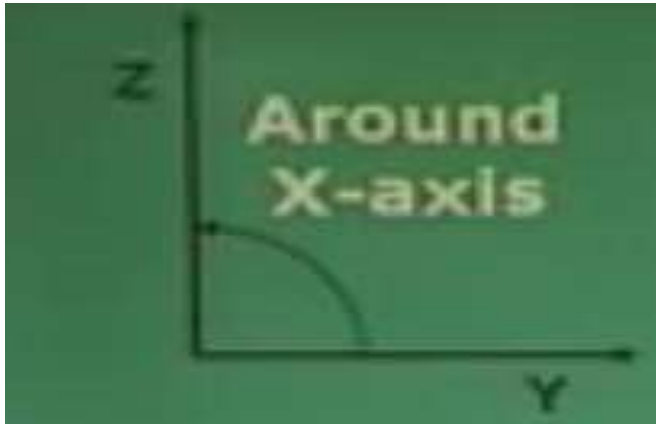
Y-axis

$$\begin{bmatrix} \cos(\gamma) & -\sin(\gamma) & 0 & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Z-axis

Why is the sign reversed in one case ?

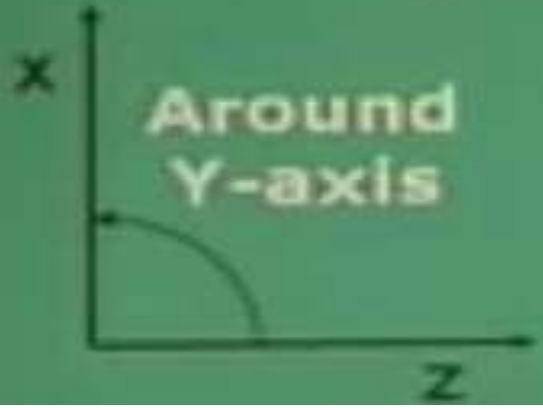

Y
X
Z
Y
Around
Z-axis
X
Z
Around
X-axis
Y
X
Around
Y-axis



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) & 0 \\ 0 & \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} \cos(\gamma) & -\sin(\gamma) & 0 & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation About an Arbitrary Axis in Space

Assume, we want to perform a rotation θ degrees, about an axis in space passi through the point (x_o, y_o, z_o) with directi cosines (c_x, c_y, c_z).

1. First of all, translate by:
$$|T| = -(x_o, y_o, z_o)^T$$
2. Next, we rotate the axis into one of the principle axes, let's pick, Z ($|R_x|$, $|R_y|$).
3. We rotate next by θ degrees in Z ($|R_z(\theta)|$
4. Then we undo the rotations to align the axis.
5. We undo the translation: translate by $(-x_o, -y_o, -z_o)^T$

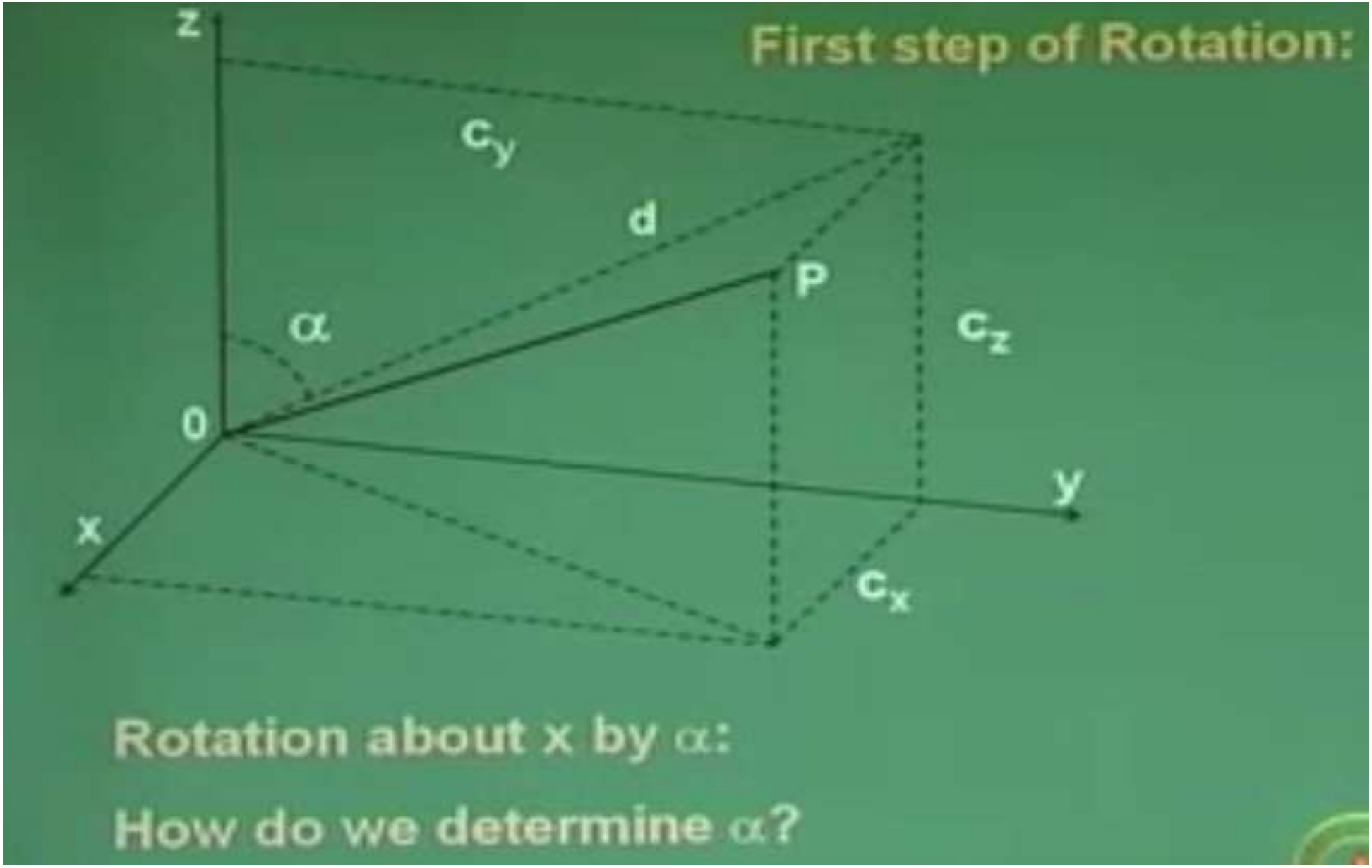
The tricky part of the algorithm is in step(2), as given before.

This is going to take 2 rotations:

i) About x-axis
(to place the axis in the xz plane)

and

ii) About y-axis
(to place the result coincident with the z-axis).


First step of Rotation:
z
c_y
d
P
α
c_z
0
y
x
c_x
Rotation about x by α:
How do we determine α?

Project the unit vector, along OP, into the yz plane.

The y and z components, c_y and c_z, are the direction cosines of the unit vector along the arbitrary axis.

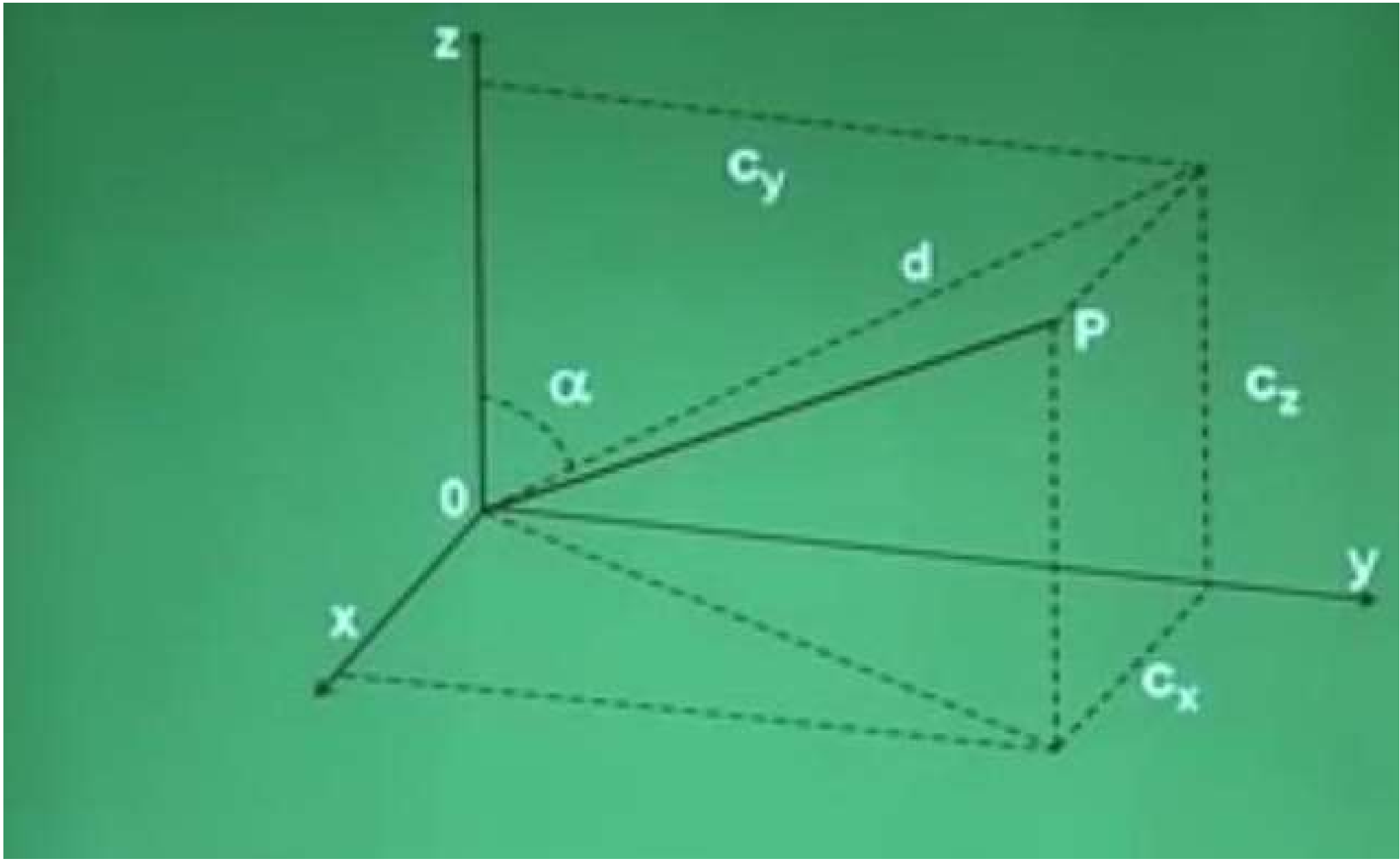
It can be seen from the diagram, that :

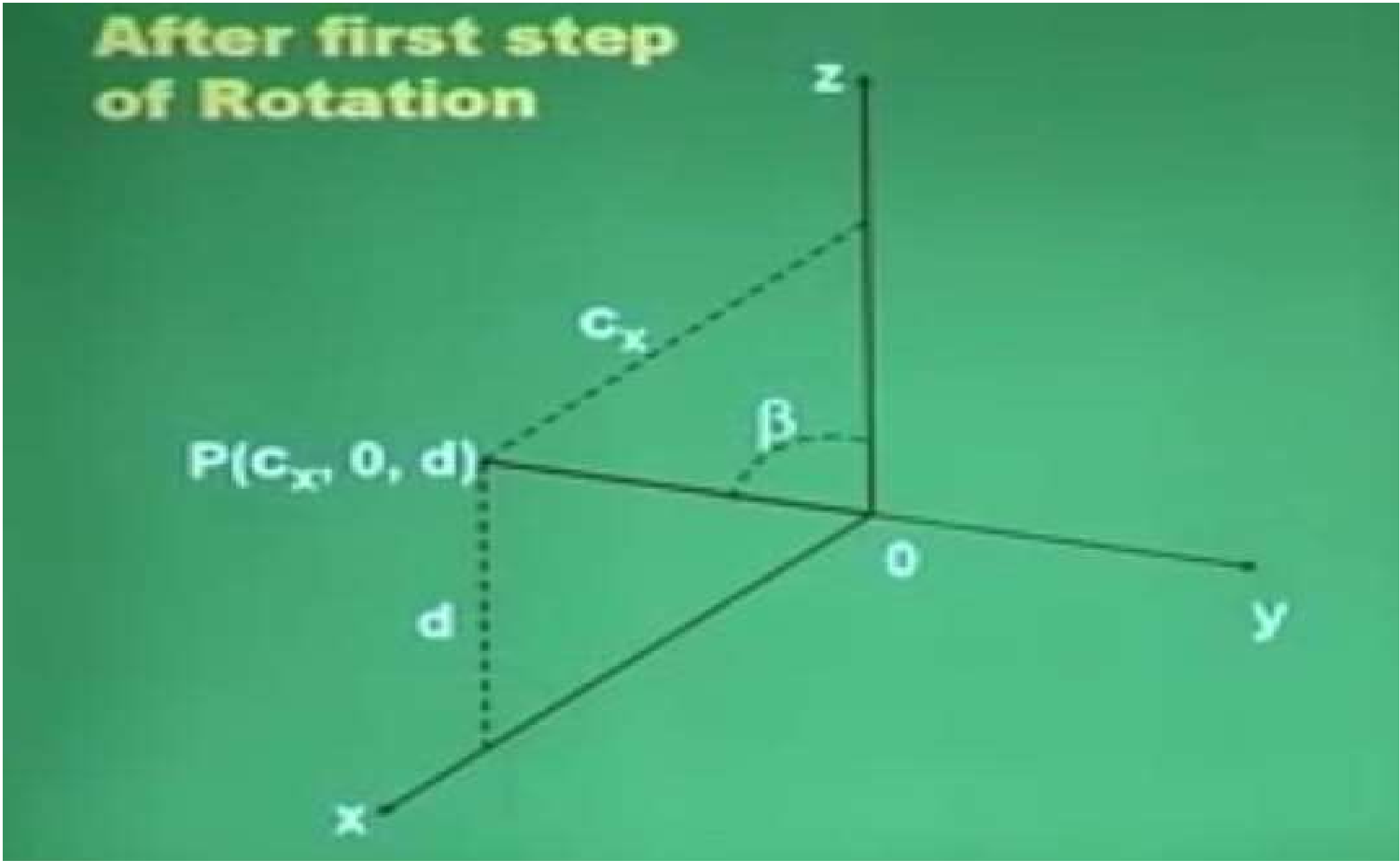
$$d = \mathrm{sqrt}\left(C_y^2 + C_z^2\right)$$

$$\cos(\alpha) = C_z / d$$

$$\sin(\alpha) = C_y / d$$

$$\alpha = \sin^{-1}\left[\frac{c_y}{\sqrt{c_y^2 + c_z^2}}\right]$$


z
c_y
d
P
α
c_z
0
y
x
c_x


After first step
of Rotation
z
c_x
β
P(c_x, 0, d)
0
y
d
x

Rotation by β about y:

How do we determine β?

Steps are similar to that done for α:

- Determine the angle β to rotate the result into the Z axis:
- The x component is c_x and the z component is d.

$$\cos(\beta) = d = d/(\text{length of the unit vector})$$

$$\sin(\beta) = c_x = c_x/(\text{length of the unit vector})$$

Final Transformation for 3D rotation, about a arbitrary axis:

$$M = |T|\,|R_x|\,|R_y|\,|R_z|\,|R_y|^{-1}\,|R_x|^{-1}\,|T|^{-1}$$

Final Transformation matrix for 3D rotation, about an arbitrary axis:

$$M = |T|\,|R_x|\,|R_y|\,|R_z|\,|R_y|^{-1}\,|R_x|^{-1}\,|T|^{-1}$$

where:

$$T = \begin{bmatrix} 1 & 0 & 0 & -x_0 \\ 0 & 1 & 0 & -y_0 \\ 0 & 0 & 1 & -z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix};$$

$$R_x = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & C_z/d & -C_y/d & 0 \\ 0 & C_y/d & C_z/d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix};$$

$$R_y = \begin{bmatrix} d & 0 & -C_x & 0 \\ 0 & 1 & 0 & 0 \\ C_x & 0 & d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix};$$

$$R_z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix};$$